RISC-V SECURE IoT GATEWAY SoC

ARCHITECTURE SPECIFICATION & REGISTER MAP
Updated for 3×12 AXI4 Interconnect, DMA, SPI & GPIO Integration

1. System Architecture & Overview

The Secure IoT Gateway SoC is organized around a 32-bit RISC-V processor (VeeR EL2) and an AXI4 system crossbar. The system is designed for secure, autonomous IoT edge sensor acquisition. The SoC integrates four primary peripheral blocks (I2C Master, SPI Master, 32-bit GPIO, and AXI-Lite UART), an autonomous DMA engine, a hardware AES-128 cryptographic accelerator, and dedicated on-chip Program and Data SRAMs.

The data path is architected to minimize CPU load: sensor measurements acquired over I2C or SPI can be streamed by the DMA engine directly into Data Memory. The processor or DMA initiates hardware AES-128 encryption on the raw blocks. The resulting ciphertext is then streamed out through the UART for encrypted uplink transmission to a cloud gateway or host system.

2. Bus Interconnect Architecture (3×12 Crossbar)

The SoC implements a 3-Master × 12-Slave AXI4 crossbar (axi_interconnect_wrap_3x12). The interconnect supports independent read and write channels, round-robin arbitration with priority support, and natural 16 MB (2^24) address window decoding per slave.

Master Port Allocation (Interconnect Slave Ports):

Master Port	Initiator Source	Width	Function / Role
s00	VeeR EL2 LSU / External BFM	32-bit	Primary CPU memory-mapped load/store unit via 64-to-32b adapter
s01	Secondary / Spare Master	32-bit	Exposed at SoC top boundary for multi-drop or external debugger
s02	Internal DMA Master	32-bit	Autonomous memory-to-peripheral block transfer master

3. SoC Memory Map (16 MB Window Spacing)

The address space is organized into 16 MB windows (2^24 bytes per slave port). High-order address bits [31:24] select the target slave:

Slave	Base Address	Range	Target Module	Description
m00	0x0000_0000	0x0000_0000 - 0x00FF_FFFF	i2c_master_top	I2C Master via axi_to_wb_bridge
m01	0x0100_0000	0x0100_0000 - 0x01FF_FFFF	axi_aes_slave	AES-128 Crypto Engine (Encrypt/Auto-Decrypt)
m02	0x0200_0000	0x0200_0000 - 0x02FF_FFFF	axi_uart_slave	UART IP Core with TX/RX FIFOs
m03	0x0300_0000	0x0300_0000 - 0x03FF_FFFF	axi_spi_top	High-speed Master SPI Controller
m04	0x0400_0000	0x0400_0000 - 0x04FF_FFFF	gpio_axi	32-bit Bidirectional GPIO with IRQ
m05	0x0500_0000	0x0500_0000 - 0x05FF_FFFF	axi_ram (Prog)	32 KB Synchronous Program SRAM
m06	0x0600_0000	0x0600_0000 - 0x06FF_FFFF	axi_ram (Data)	32 KB Synchronous Data/Payload SRAM
m07..10	0x0700_0000..	0x0700_0000 - 0x0AFF_FFFF	dummy_axi_slave	Reserved slots (returns 0xDEAD_000x)
m11	0x0B00_0000	0x0B00_0000 - 0x0BFF_FFFF	dma_axi_wrapper_v	DMA CSR & Descriptor Engine

4. Peripheral Register Reference

IP	Base	Reg	Off	Access	Function & Description
I2C	0x0000_0000	PRER_LO	0x00	R/W	Clock prescaler low byte
I2C	0x0000_0000	CTR	0x08	R/W	[7]=EN (Enable core), [6]=IEN (Interrupt enable)
I2C	0x0000_0000	TXR/RXR	0x0C	W/R	Transmit Data (write) / Receive Data (read)

I2C	0x0000_0000	CR/SR	0x10	W/R	Command (STA, STO, RD, WR) / Status (BUSY, TIP, ACK)
AES	0x0100_0000	CTRL	0x00	R/W	[0]=LD (Encrypt start), [1]=KLD, [16]=DONE, [17]=KDONE
AES	0x0100_0000	KEY0..3	0x04..10	R/W	128-bit Encryption Key (32-bit words, MSB first)
AES	0x0100_0000	TEXT0..3	0x14..20	R/W	128-bit Plaintext Input Block (MSB first)
AES	0x0100_0000	OUT0..3	0x24..30	RO	128-bit Ciphertext Output Block
AES	0x0100_0000	DOUT0..3	0x34..40	RO	128-bit Recovered Plaintext from auto-chained decrypt
UART	0x0200_0000	THR/RBR	0x00	W/R	TX FIFO Push (write) / RX FIFO Pop (read) (DLAB=0)
UART	0x0200_0000	IER	0x04	R/W	Interrupt Enable ([0]=RX Data Available, [1]=THRE)
UART	0x0200_0000	BAUD	0x08	R/W	Baud Divisor register = clk/(16*baud) (DLAB=1)
UART	0x0200_0000	LCR	0x0C	R/W	Line Control: [7]=DLAB, [1:0]=8N1 configuration
UART	0x0200_0000	LSR	0x14	RO	Line Status: [6]=TEMT, [5]=THRE, [0]=DATA_READY
SPI	0x0300_0000	CR/SR	0x60/64	R/W	SPI Control (Enable, Master, CPOL, CPHA) & Status
SPI	0x0300_0000	DTR/DRR	0x68/6C	W/R	SPI Data Transmit (write) & Data Receive (read)
GPIO	0x0400_0000	DATA/DIR	0x04/08	R/W	Output data drive [31:0] and Direction [31:0] (1=Out)
DMA	0x0B00_0000	CONTROL	0x00	R/W	[0]=GO, [1]=ABORT, [7:4]=MAX_BURST
DMA	0x0B00_0000	STATUS	0x08	RO	[0]=BUSY, [1]=DONE, [2]=ERROR
DMA	0x0B00_0000	DESC0	0x20..38	R/W	SRC_ADDR, DST_ADDR, NUM_BYTES, and CFG registers